Ikjoo Oh

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EDUCATION

University of Wisconsin-Madison (UW-Madison) | B.S. in Computer Science and Data Science *Aug. 2024 - Present*
Currently attending in good academic standing. GPA 3.82/4.0

George Mason University (GMU) | B.S. in Data Science *Sept. 2021 - May. 2024*
Transfer

PUBLICATIONS & PREPRINTS

[1] Evaluation of RAG-Based Approach for SNOMED CT Concept Mapping of Multi-Institutional Local Clinical Terminology
Chansik Kim, **Ikjoo Oh**, Jiho Kim, Taehoon Ko
American Medical Informatics Association (AMIA, Q1 Journal) 2026 / Second Author

[2] Predicting comorbid chronic diseases using multi-label learning; a comparative study with electronic health records
Kanghyuck Lee, **Ikjoo Oh**, Eunsik Choi, Sung-Hwan Kim, Taehoon Ko

RESEARCH / INTERNSHIP EXPERIENCE

The Catholic University of Korea, Medical Informatics Lab *Jan. 2025 - May 2026*
Undergraduate Research Intern (Advisor: Taehoon Ko)

- Developed a RAG-based framework for mapping multi-institutional local clinical terminology to SNOMED CT, RxNorm, and LOINC.
- Improved baseline mapping accuracy from 60% to 93% by adding an LLM-based candidate selection stage and manually curating and refining the provided dataset.

CNAPS.AI *Sep. 2026 - Present*
Intern, AI Orchestration Team

SELECTED PROJECTS

[1] **League of Legends Minimap Object Detection and Game Analytics Platform**

- Independently developed an end-to-end YOLO-based object detection pipeline for identifying wards and other in-game events from League of Legends minimap footage.
- Addressed limited and noisy real-world training data through three rounds of dataset refinement and synthetic data generation, improving detection accuracy from approximately 30-40% to 90%.
- Integrated the detection pipeline with gameplay analytics, UI/UX, and database components to provide automated post-game analysis.

[2] **Ovarian Cancer Image Analysis Using CNNs**

- Used U-Net as a baseline, then applied a denoising autoencoder to reduce image noise, improving accuracy by approximately 15%.
- Investigated the challenges of learning from heterogeneous medical imaging modalities, including ultrasound, CT, and MRI.

[3] **Bank Customer Retention Strategy Using Reinforcement Learning**

- Developed a policy-based reinforcement learning prototype to explore customer-retention decisions using a simulated banking dataset.
- Designed and evaluated the reward structure and learned policy rather than optimizing conventional classification accuracy.
- Built an interactive Streamlit application to visualize customer states, policy decisions, and retention outcomes.

AWARDS AND HONORS

Ideathon Finalist, LIKELION:US *Fall 2024*
Advanced to the final round of a startup idea competition hosted by LIKELION:US.

Dean's List, UW-Madison *Fall 2024*
Awarded for good academic standing.

Champion, IGC League of Legends Tournament, representing GMU *Fall 2023*

Champion, Chungju World Martial Arts Festival Cup Esports Tournament *2019*

Runner-Up, KeG Presidential Amateur Esports Competition, Incheon Regional Qualifier *2019*

LEADERSHIP & ACTIVITIES

Data Science Lead, LIKELION Coding Club, UW-Madison *Fall 2024, Spring 2026, Fall 2026*

Undergraduate Peer Mentor, UW-Madison *Spring 2026 - Present*
Mentored five undergraduate students through ongoing academic and technical guidance.

Member, Coder:US Coding Club, George Mason University *Spring 2021, 2024*

Varsity League of Legends Team Captain, UW-Madison Esports *Fall 2024*

Varsity Tennis Team Member, George Mason University *Fall 2023*

Second-Team Sabre Fencer, UW-Madison Fencing Club *Fall 2024*

TECHNICAL SKILLS

Programming Languages
Python, Java, C, SQL, Linux

Cloud Computing
Docker, Amazon Web Services

Machine Learning & Data Science
TensorFlow, PyTorch, Keras, scikit-learn, pandas, NumPy, matplotlib, seaborn

MISCELLANEOUS

Military Interpreter to the U.S. Army, Ground Operations Command (GOC) *Feb. 2023 - Mar. 2023*
Provided Korean-English interpretation and operational support during Freedom Shield, a joint ROK-U.S. military exercise.